

Activity Dossier

Lab 4 — Forecasting: How Many Riders Next Week?

AI & Data Science with GenAI (Rail) · Advanced ML — your forecasting lab

Student package — everything you need to do this activity

Field	Detail
Module	Advanced ML — Forecasting
Builds on	Regression Models, Model Validation, and Lab 2 (train/test by time, honest baselines)
Duration	~2 hours
You need	A browser + JupyterLab (or Google Colab)
Data	Real MTA daily ridership, 1,339 days (Jan 2023 – Aug 2026)
Hand in	Your completed notebook + a written report

1. The situation

You have joined the operations planning team of the Long Island Rail Road (LIRR) — one of the busiest commuter railroads in the world, carrying around a quarter of a million people on a normal weekday.

Every week the team must publish next week's train and crew plan. To do that they need a daily ridership forecast for the seven days ahead: which days need extra cars, which need fewer, and how many crews to roster.

In Lab 2 you built a detector — it looked at today's sensors and said "this is a failure". Today you predict a number for a day that has not happened yet. That is a different and harder job, and it comes with one rule that shapes everything:

The 7-day rule. You publish the plan a week in advance, so when you forecast next Tuesday you do not know Monday's numbers — you know nothing from the last seven days. Every feature you build must be at least 7 days old. Break that rule and you build a model that looks brilliant in the notebook and is useless in the depot.

2. What you will be able to do

- Prepare a real time series: check the calendar is complete, and repair impossible values without breaking the row-to-day relationship.
- Split a time series into train and test by time before exploring it, and explain what a 92-day test can and cannot prove.
- Find what repeats — measure weekly and yearly seasonality on the training period, and say which one matters more.
- Build three forecasting baselines (naive, seasonal naive, flat) and score them.
- Score a numeric forecast with MAE, RMSE and MAPE — and say which one belongs in a management report.
- Engineer lag, calendar and trend features and fit a LinearRegression forecast — your one new idea this lab.
- Judge honestly whether the model beat the one-line rule, and diagnose where it fails.

3. What's in your package

File	What it is
lab4_forecasting.ipynb	The notebook you work in — open this.
lirr_daily_ridership.csv	The real dataset (already provided).
Student_Report_Lab4.docx	Your written report — the main deliverable.
Activity_Dossier_Lab4.pdf	This document — your guide to the activity.
README_Lab4_data.md	Where the data comes from, and its terms of use.

File	What it is
START_HERE.txt	One-page quick start.

4. Before you start

Keep the notebook and the CSV in the same folder, open the notebook, and run cells from the top with Shift+Enter.

```
# In JupyterLab: open a terminal in the notebook folder, then
jupyter lab
# then open lab4_forecasting.ipynb and run each cell top to bottom
```

In Google Colab: upload the notebook, then use the Files panel on the left to upload lirr_daily_ridership.csv into the session before running Step 0.

How the notebook works: every step is fully worked and explained — read it, run it, then do the  **Your turn** cell yourself. There are nine of them, and the lab ends with a two-part  **Challenge**. Only one idea is new at a time.

5. The activity, step by step

Step	What you do	Already learned / New?
0-1	Load the series, check the calendar is complete, plot it	Already learned
2	Data preparation: find impossible values and repair them	NEW — repairing a time series
3	Split train / test by time at 1 June 2026 — before exploring	Already learned (Lab 2)
4	Find the rhythm on the TRAINING data: weekly and monthly seasonality	Already learned
5	Three baselines: naive, seasonal naive, flat	NEW — baselines before models
6	Score forecasts with MAE, RMSE and MAPE	NEW — regression metrics
7	Engineer lag / calendar / trend features, fit LinearRegression	NEW — feature engineering
8	Did the model beat the one-line rule?	Already learned (Lab 2 habit)
9	Look at what it got wrong, day by day	Already learned
	Challenge: a holiday feature, and the price of being wrong	Your turn

6. What to hand in

Deliverable	Notes
Completed notebook	All steps run top to bottom; the nine  Your turn cells filled; the  Challenge attempted.
Written report	Student_Report_Lab4.docx, in your own words. Quote a number as evidence where it helps — but a report that is only pasted output scores no marks.
Show & tell	Be ready to show one chart and state one thing your model gets wrong.

7. How you are assessed (100 marks)

Component	Marks	Full marks means
Worked steps run +  Your turn cells done	30	Notebook runs top to bottom; all nine practice cells completed with a written interpretation, not just code
 Challenge (both parts)	20	Holiday model retrained and scored; cost function written and the cheapest safety margin found
Baselines vs model, argued honestly	20	Quotes the seasonal-naïve score, states the model's gain, and says plainly whether it justified the work
Failure analysis and limits	20	Names where the model breaks and why, and states the limits of a 92-day summer-only test
A clear recommendation	10	Says what you would actually deploy next week, and what you would build next

8. Your data — where it comes from

The dataset is real and official: daily ridership and traffic estimates published by the Metropolitan Transportation Authority (MTA) on the New York State open data portal, data.ny.gov (dataset "MTA Daily Ridership and Traffic: Beginning 2020", id sayj-mze2).

It is released under the OPEN-NY Terms of Use, which are among the least restrictive open-data terms in use: free to download and reuse, in public or private projects, with no attribution or share-alike requirement — the only condition being that your use is lawful. We credit the MTA anyway, because that is good practice.

Your package contains the three rail columns, cut to 1 January 2023 – 31 August 2026. The values are exactly as published — including four days that report zero LIRR riders, which you diagnose and repair yourself in Step 2. Two of them turn out to be different kinds of day, and the other two columns are what tell you so.

Column	Meaning
date	The calendar day (one row per day, no gaps)
lirr_riders	Long Island Rail Road — estimated riders that day. This is what you forecast.
mnrr_riders	Metro-North Railroad — a second commuter railroad, for cross-checking odd days
subway_riders	New York City Subway — for cross-checking odd days

A caveat the MTA states itself: these are next-day estimates, published fast, and they can differ from the official ridership totals the MTA publishes later. That is normal for operational data — and it is exactly the kind of thing a good report mentions.

9. Getting unstuck

- Run cells from the top, in order — later cells depend on variables created earlier.
- `FileNotFoundError` means the CSV is not beside the notebook. Unzip the package properly, or re-upload the CSV in Colab.
- Never sort, shuffle or delete rows in a time series. Everything in this lab depends on "7 rows back = 7 days back".
- If a score looks too good, check whether a feature is newer than 7 days. That is leakage, and it is the classic forecasting mistake — and it applies to baselines and to any threshold or margin you tune, not just to model features.
- An error of 20,000 riders is not the same thing on a Sunday as on a Wednesday. Always ask "per cent of what?".
- A chart without a title and axis labels is not finished.

Stuck? Re-read the step's Insight box, then ask a teaching assistant.